

Indoor Visible Light Positioning with Angle Diversity Transmitter

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Abstract—A new concept for indoor positioning using visible light communication (VLC) is presented in this paper. The feasibility of uplink localization is verified so that the proposed system can work in conjunction with downlink positioning systems to improve overall localization accuracy. We propose to use an angle diversity transmitter (ADT) associated with accelerometers for uplink three-dimensional localization. Unlike other techniques using VLC for positioning, the proposed system can achieve indoor localization without making assumptions about the height or orientation angle of a mobile user. The received signal strength (RSS) method is used to minimize implementation costs while achieving satisfactory positioning accuracy. Simulation results show that an average localization error of less than 0.15 m can be achieved even when the receiver is tilted by 45° . Also, compared with a single light emitting diode (LED), the ADT is found to be more robust to accelerometer measurement errors.

I. INTRODUCTION

The demand for indoor positioning and navigation systems is continuously increasing due to the expansion of location-based services. Because of strong attenuation and shadowing effects, global positioning systems (GPS) do not work in indoor environments. Therefore, new indoor positioning systems using different techniques are required. A number of indoor positioning systems have been proposed and studied using various wireless technologies, such as radio frequency identification (RFID), infrared (IR), ultrasonic and wireless local area network (WLAN). As an alternative, indoor positioning systems using visible light communication (VLC) have recently attracted much attention in both academia and industry [1], [2]. The widespread use of light emitting diodes (LEDs) for illumination offers new opportunities for indoor wireless communication. Compared with incandescent light bulbs, LEDs have lower energy consumption and a longer lifetime. Also, the absence of interference with radio frequency (RF) communication systems makes VLC particularly promising in electromagnetic sensitive areas, such as aircraft cabins, hospitals and intrinsically safe environments.

Current techniques used for indoor visible light positioning (VLP) are mostly based on triangulation, trilateration and fingerprinting. Triangulation uses relative angles to a reference point for positioning. A typical example for triangulation is the angle of arrival (AOA) method [3], [4]. Trilateration requires

several measurements of the distance to a reference point for positioning. This can be achieved by methods such as time of arrival (TOA), time difference of arrival (TDOA) and received signal strength (RSS) [3], [4]. The TOA method is the basis of GPS systems and it requires transmitted signals to be precisely synchronized. This is however not viable for low-cost indoor positioning systems. As for the TDOA method, localization information is extracted from the difference in the path lengths of two or more signals impinging on multiple receivers. Although precise synchronization is not needed for TDOA, it requires at least two receivers with their locations known to perform localization. Also, as the optical signal travels at the speed of light and the difference in the path lengths in indoor environments is generally small, extremely accurate time measurements are required for the TDOA method.

VLP systems are generally based on intensity modulation and direct detection (IM/DD), and they can achieve high positioning accuracy using the RSS method. Indoor positioning systems achieving centimeter-level accuracy have been reported [1], [5], [6], assuming both the height and orientation angle of the mobile station (MS) are fixed and known. In practice, however, the orientation of a MS is random and may change continuously. Although the orientation of a MS can be measured by sensors such as accelerometers, which are already implemented in modern smartphones, there may be situations where trilateration using a single LED becomes infeasible because the optical transmitter(s) and receiver(s) of interest are not well-aligned. Therefore, new positioning systems without such assumptions are of great interest. For example, decimeter-level indoor positioning using unmodified smartphones has recently been reported in [7]. The drawback of this method is the requirement of complicated image processing techniques and cloudlet servers, which results in low energy and cost efficiencies.

The aim of this study is to develop an uplink indoor positioning system so that it can work in conjunction with downlink positioning systems to achieve increased localization accuracy. Localization errors of the proposed system are analyzed with respect to the height and tilt angle of the mobile user. The imperfection of accelerometer measurements is also included in the performance analysis.

The rest of the paper is organized as follows. System model and notations are introduced in Section II. An algorithm using the RSS method for uplink three-dimensional indoor localization is proposed in Section III. Positioning errors caused by system noise and accelerometer measurement errors are also analyzed. Simulation results are presented and discussed in Section IV. Finally, conclusions are given in Section V.

II. SYSTEM MODEL AND NOTATIONS

For indoor positioning systems using the RSS method, multiple transmitters and/or receivers are used for trilateration. For the proposed uplink positioning system, LEDs implemented on a MS constitute the optical transmitter, and photodiodes (PDs) distributed in the ceiling are the receivers. The system configuration is shown in Fig. 1. The LED source can be described by a set of parameters, denoted by $\mathcal{S} = \{\mathbf{r}_t, \mathbf{n}_t, m, P_t\}$, where $\mathbf{r}_t = [x_t, y_t, z_t]^T$ represents the position of the LED, with $[\cdot]^T$ denoting the transpose of a matrix; $\mathbf{n}_t$ is the unity normal vector of the LED; m is the Lambertian order of the LED; and P_t is the transmit power of the LED. Similarly, the k -th PD in the ceiling is represented by set $\mathcal{R}_k = \{\mathbf{r}_{r,k}, \mathbf{n}_{r,k}, A_k, \Psi_{c,k}\}$, $\forall k \in \{1, 2, \dots, L\}$, where $\mathbf{r}_{r,k} = [x_{r,k}, y_{r,k}, z_{r,k}]^T$ represents the location of the k -th PD; $\mathbf{n}_{r,k}$ represents the unity normal vector of the k -th PD; A_k is the physical area of the k -th PD; $\Psi_{c,k}$ is the field of view (FOV) of the k -th PD; and L is the total number of PDs in the ceiling. It is assumed that all the parameters of the L PDs are the same except their geographical locations. Therefore, $\mathcal{R}_k$ can be simplified to $\mathcal{R}_k = \{\mathbf{r}_{r,k}, \mathbf{n}_r, A, \Psi_c\}$, and all of the PDs are assumed to face vertically downwards to the floor, i.e., $\mathbf{n}_r = [0, 0, -1]^T$.

A. Optical Channel

Although the VLC channel includes both line-of-sight (LOS) and diffuse components, it has been reported that the energy of the reflected signal is much lower than the energy of the LOS component [8]. For simplicity, only LOS links are considered in this paper. Assuming the LED of interest follows the Lambertian radiation pattern [9]–[11], the direct current (DC) gain of the LOS optical channel between the LED and k -th PD is given by:

$$H_k = \frac{(m+1)A}{2\pi d_k^2} \cos^m(\phi_k) T(\psi_k) g(\psi_k) \cos(\psi_k), \quad (1)$$

where d_k is the distance between the LED and the k -th PD; ϕ_k is the irradiance angle with respect to the normal vector of the LED; ψ_k is the incidence angle of light with respect to normal vector of the k -th PD; and $T(\psi_k)$ is the gain of the optical filter used at the k -th PD. The Lambertian order m of the LED is calculated by $m = -\frac{1}{\log_2(\cos(\Phi_{1/2}))}$, with $\Phi_{1/2}$ denoting the semi-angle of the LED. The gain of the optical concentrator at the k -th PD is denoted by $g(\psi_k)$, and it is calculated as [9]:

$$g(\psi_k) = \begin{cases} \frac{n^2}{\sin^2(\Psi_c)}, & 0 \leq \psi_k \leq \Psi_c \\ 0, & \psi_k > \Psi_c \end{cases}, \quad (2)$$

where n is the refractive index of the optical concentrator.

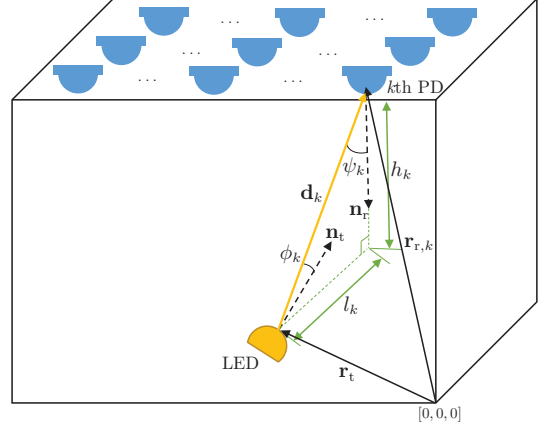


Fig. 1. System configuration.

B. RSS and Noise Analysis

The desired optical power at the k -th PD, denoted by $P_{\text{opt},k}$, is calculated as:

$$P_{\text{opt},k} = P_t H_k. \quad (3)$$

After optical to electrical (O/E) conversion, the measured electrical power at the k -th PD is given by:

$$P_{\text{elec},k} = (R_p P_{\text{opt},k})^2, \quad (4)$$

where R_p represents the responsivity of the PD. The measured electrical power is affected by both shot noise and thermal noise in the PD. The power of shot noise is given by [12]:

$$\sigma_{\text{shot},k}^2 = 2qR_p P_{\text{opt},k} B + 2qI_{\text{BG}} I_2 B, \quad (5)$$

where q is the elementary charge; B is the equivalent noise bandwidth; I_{BG} is the background current; and I_2 is a noise bandwidth factor. The power of thermal noise is given by [12]:

$$\sigma_{\text{thermal}}^2 = \frac{8\pi k_B T_K}{G} \eta A I_2 B^2 + \frac{16\pi^2 k_B T_K \Gamma}{g_m} \eta^2 A^2 I_3 B^3, \quad (6)$$

where k_B is the Boltzmann constant; η is the field capacitance per unit area of the PD; T_K is the absolute temperature of the circuit; G is the open-loop voltage gain; g_m is the transconductance of the field-effect transistor (FET) used at the front end of the receiver; Γ is the FET channel noise factor; and I_3 is an additional noise bandwidth factor. The signal-to-noise ratio (SNR) is defined as:

$$\text{SNR}_k = \frac{(R_p P_t H_k)^2}{\sigma_{\text{shot},k}^2 + \sigma_{\text{thermal}}^2}. \quad (7)$$

III. LOCALIZATION ALGORITHM

A. Angle Diversity Transmitter

Following [6], where an angle diversity receiver (ADR) and a single LED array are used for trilateration, in this paper we propose an uplink positioning system using an angle diversity transmitter (ADT) in conjunction with accelerometers at the MS. Two types of ADT are considered: ADT with 19 LEDs,

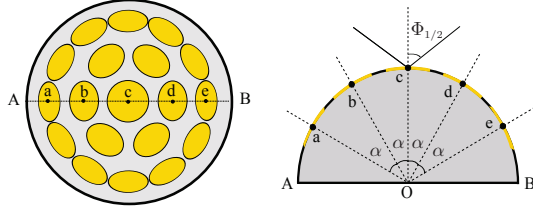


Fig. 2. An ADT with 19 LEDs (ADT-19).

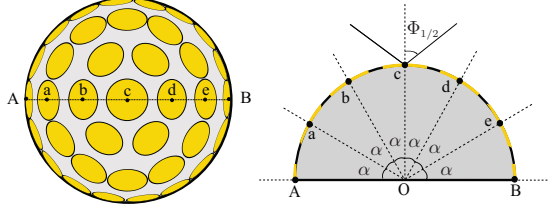


Fig. 3. An ADT with 37 LEDs (ADT-37).

and ADT with 37 LEDs, which are abbreviated as ADT-19 and ADT-37, respectively. The layout of ADT-19 is shown in Fig. 2, where 19 LEDs with the identical characteristics are mounted on the surface of the semi-sphere. One LED is mounted at the center of the surface while the rest are placed symmetrically along the side. The LEDs of an ADT are grouped into different branches. The center LED forms the first branch. The second branch consists of six LEDs evenly located on a circle and displaced by a polar angle α , and the third branch contains twelve LEDs on a circle with a polar angle 2α . Similarly, the layout of ADT-37 is illustrated in Fig. 3. It is formed in a similar way as ADT-19, with an additional fourth branch of eighteen LEDs displaced by a polar angle 3α . It is assumed that the normal vectors of all the LEDs are known *a priori*, as these are determined by the physical design of the ADT. In an ADT, each LED is switched on in sequence and the optimum LED which exhibits the highest channel gain is selected. This technique diminishes the dependence of the positioning accuracy on the orientation of the MS, but it requires a closed-loop communication link. The time required for this adaptive LED selection process is assumed to be small and hence negligible.

B. Three-Dimensional Localization

1) *Single LED Transmitter*: Assume the orientation vector of the tilted MS lies in the yz plane, with θ denoting the clockwise displacement angle with respect to the z axis. The normal vector of the LED measured by the accelerometer can be expressed as:

$$\mathbf{n}_t = [0, \sin(\theta'), \cos(\theta')]^T, \quad (8)$$

where $\theta' = \theta + \theta_{\text{err}}$ is the measured tilt angle of the MS, with θ_{err} representing the accelerometer measurement error. Combining (1)–(3) and (8), the received optical power at the

TABLE I
SIMULATION PARAMETERS

Parameter name, notation	Value
ADT polar angle, α	30°
LED transmit power, P_t	2.2 W
LED semi-angle, $\Phi_{1/2}$	60°
FOV of the PD, Ψ_c	90°
Surface area of the PD, A	1 cm ²
PD responsivity, R_p	0.5 A/W
Reflective index, n	1.5
Optical filter gain, T	1
Equivalent noise bandwidth, B	100 Mb/s
Background current, I_{BG}	5100 μ A
Noise factor I_2	0.562
Noise factor I_3	0.0868
Circuit absolute temperature, T_K	295 K
Open-loop gain, G	10
FET Transconductance, g_m	30 mS
FET channel noise factor, Γ	1.5
Field capacitance per unit area, η	112 pF/cm ²

k -th PD can be written as:

$$P_{\text{opt},k} = \frac{C(z_{r,k} - z_t) \left((y_{r,k} - y_t) \sin(\theta') + (z_{r,k} - z_t) \cos(\theta') \right)^m}{\left(\sqrt{(x_{r,k} - x_t)^2 + (y_{r,k} - y_t)^2 + (z_{r,k} - z_t)^2} \right)^{m+3}}, \quad (9)$$

where $C = \frac{P_t(m+1)A}{2\pi} T(\psi_k)g(\psi_k)$ is a constant, which is determined by the physical parameters of the LED and PDs. In (9), the arguments are the coordinates of the MS, i.e., x_t , y_t and z_t . Therefore, at least three measurements of the received power at the PDs are required for three-dimensional localization. Considering a system of three nonlinear equations having the form of (9), a closed-form solution is not straightforward. Therefore, we resort to exhaustive search to find the coordinates of the MS.

2) *Angle Diversity Transmitter*: For an ADT, the LED achieving the highest channel gain is selected for determining the location information of the MS. The unity normal vector of the w -th LED, assuming no tilt at the MS, is denoted by $\mathbf{n}_{t,w} = [e_{x,w}, e_{y,w}, e_{z,w}]^T$. After rotating the MS clockwise in the yz plane by an angle θ' , the measured normal vector of the w -th LED is given by:

$$\mathbf{n}'_{t,w} = \mathbf{R} \mathbf{n}_{t,w}, \quad (10)$$

where

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta') & \sin(\theta') \\ 0 & -\sin(\theta') & \cos(\theta') \end{bmatrix}. \quad (11)$$

Combining (1)–(3), (10) and (11) gives the received optical power at the k -th PD from the w -th LED in the ADT, and the expression is shown in (12).

IV. SIMULATION RESULTS

A room of size of 6 m $\times$ 8 m $\times$ 4 m is considered. The parameters used for the simulation are summarized in

$$P_{\text{opt},k,w} = \frac{C(z_{r,k} - z_t)}{\left(\sqrt{(x_{r,k} - x_t)^2 + (y_{r,k} - y_t)^2 + (z_{r,k} - z_t)^2} \right)^{m+3}} \times \left((x_{r,k} - x_t)e_{x,w} + (y_{r,k} - y_t)(e_{y,w} \cos(\theta') + e_{z,w} \sin(\theta')) + (z_{r,k} - z_t)(-e_{y,w} \sin(\theta') + e_{z,w} \cos(\theta')) \right)^m. \quad (12)$$

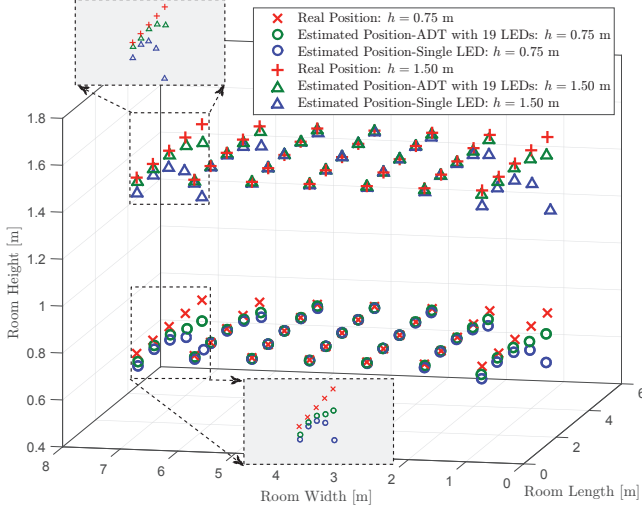


Fig. 4. Three-dimensional localization of a non-tilted MS without accelerometer measurement errors: comparing a single LED with ADT-19.

Table I. Theoretically, given three measurements of the received electrical power at the PDs, the position of the MS can be estimated. However, the estimation accuracy is affected by shot noise, thermal noise and the accelerometer measurement errors in the system. In the simulation setup, three fixed PDs are considered. They are located in the ceiling with coordinates of $[2, 3, 4]^T$, $[2, 5, 4]^T$ and $[4, 4, 4]^T$, respectively. Note that the locations of the three PDs are chosen as a simple example, which do not necessarily give the optimum localization performance. Optimizing the locations of the PDs can further improve the positioning accuracy of the system, which is subject of future work.

In Fig. 4, localization is performed at two different heights: 0.75 m (desktop height), and 1.50 m (human height). It can be seen that localization is more accurate at the desktop level, where higher SNR is achieved due to smaller irradiance and incidence angles. Compared with a single LED transmitter, it is shown that ADT-19 can achieve higher positioning accuracies, especially at the locations where the optical transmitter and receiver are not well-aligned, i.e., room corners.

Fig. 5 shows the localization of a non-tilted MS with imperfect accelerometer measurements. It is shown that inaccurate measurements of the tilt angle at the MS will degrade localization accuracies for both ADT-19 and the single LED transmitter. However, compared with the single LED trans-

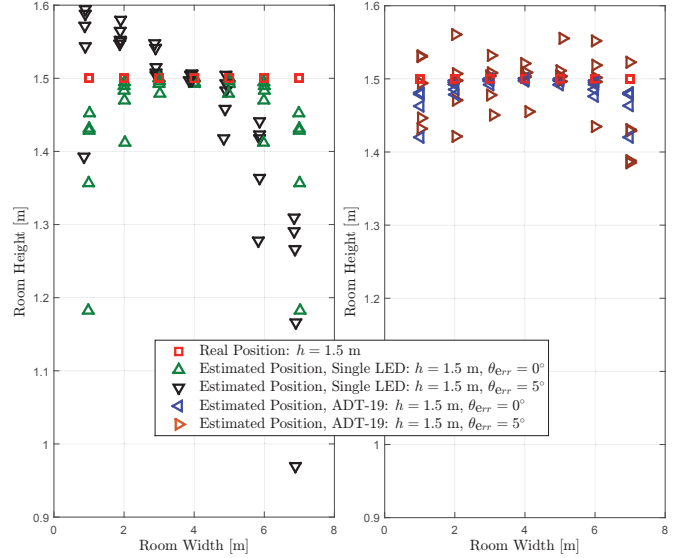


Fig. 5. Localization of a non-tilted MS with accelerometer measurement errors: comparing a single LED with ADT-19.

mitter, ADT-19 not only achieves higher positioning accuracy, but also is more robust to accelerometer measurement errors. For example, when $\theta_{\text{err}} = 5^\circ$ the maximum positioning error achieved by using a single LED is 0.531 m, which can be reduced to 0.112 m if the ADT-19 is used.

Fig. 6 shows the cumulative distribution function (CDF) of positioning errors for a single LED, ADT-19 and ADT-37. Firstly, it can be seen that for all three types of optical transmitters, localization performance degrades when the tilt angle of the MS increases. For example, a non-tilted single LED can achieve an average positioning error of 0.15 m, assuming no accelerometer measurement errors. However, if the LED is tilted by 45° the average positioning error would increase to 2.57 m. The performance degradation caused by the tilted MS can be alleviated by using the ADT. Among the aforementioned three types of transmitters, ADT-37 results in the highest robustness to MS tilt variations. It can achieve an average positioning error of 0.10 m even when the MS is tilted by 90° . Secondly, Fig. 6 also shows that the accuracy of accelerometer measurements plays a significant role in the system performance if a single LED is used. For systems using ADT-19 and ADT-37, localization becomes more robust to accelerometer measurement errors, even in the scenario when the MS is tilted by 90° .

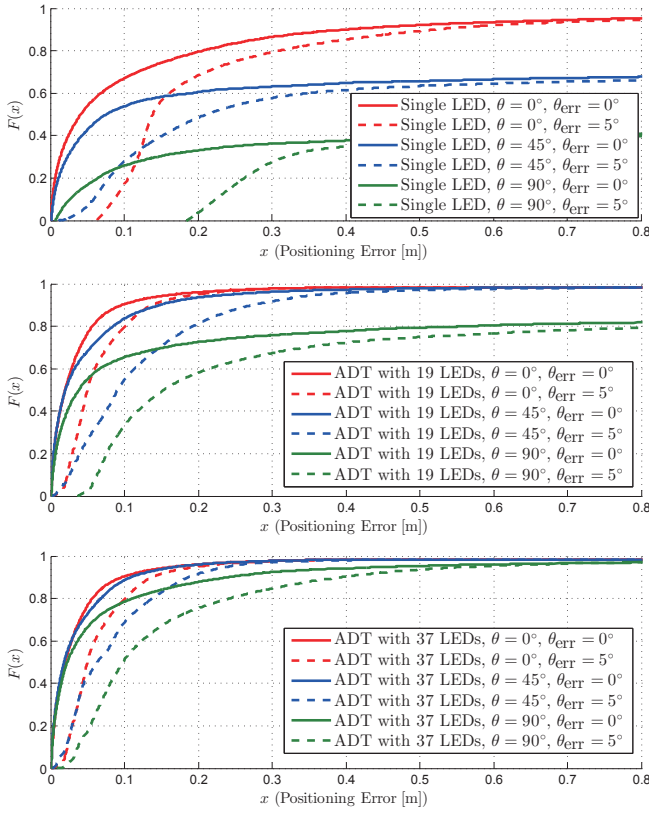


Fig. 6. CDF of positioning errors for different tilt angles at the MS ($h = 1.5$ m): comparing a single LED, ADT-19 and ADT-37.

Fig. 7 shows the localization performance degradation for different accelerometer measurement accuracies. It can be seen that when the MS is tilted by 45° , a single LED fails to perform localization even with perfect accelerometer measurements. The asymptotic-like behavior of the single LED shows that there exists a large discontinuity in the position errors. However, the ADT-39 can achieve an average positioning error of 0.15 m tolerating 10° accelerometer measurement errors.

V. CONCLUSION

We have proposed a three-dimensional uplink indoor positioning system using an ADT with distributed PDs. The proposed system uses the RSS method for localization, as well as knowledge of the orientation angle of the MS, obtained for example by integrated acceleration sensors. Using accelerometers in coordination with an ADT, indoor positioning becomes more accurate and capable of dealing with user uncertainties, such as the orientation angle. Simulation results show that, compared with a single LED transmitter, using an ADT can achieve reduced localization errors, especially in the situation where the MS of interest is greatly tilted. Also, it is shown that indoor positioning systems using an ADT are more robust to accelerometer measurement errors. Therefore, the feasibility of the proposed uplink positioning system shows great promise to be combined with downlink positioning systems to achieve enhanced localization performance.

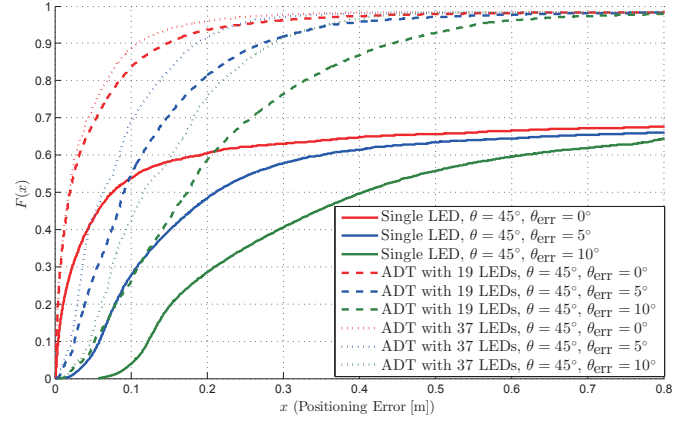


Fig. 7. CDF of positioning errors for different accelerometer measurement errors: comparing a single LED, ADT-19 and ADT-37. Note that for illustration purposes, the x -axis confined within 0.8 m, as the solid curves take much larger values to reach to probability of 1.

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